

DevOps System Design Interview Questions

50 frequently asked questions — Core Concepts, DevOps-Specific & Scenario-Based

Core System Design Concepts

1. Scalability kya hai? Horizontal vs Vertical scaling mein difference batao.
2. Load balancer kya hota hai aur kaise kaam karta hai?
3. Load balancing algorithms kaunse hain (Round Robin, Least Connections, etc.)?
4. Caching kya hai? Kab aur kahan use karte hain?
5. CDN kya hota hai aur kaise kaam karta hai?
6. SQL vs NoSQL database — kab kaunsa use karoge?
7. Database replication kya hoti hai?
8. Database sharding kya hai?
9. CAP theorem explain karo.
10. Consistency, Availability, Partition tolerance ka trade-off samjhao.
11. Microservices vs Monolithic architecture — differences aur trade-offs.
12. API Gateway kya hota hai aur kyun use karte hain?
13. Message queue kya hai? (Kafka, RabbitMQ, SQS)
14. Rate limiting kaise implement karoge?
15. Reverse proxy vs Forward proxy mein difference.

Reliability & Availability

16. High Availability (HA) kaise design karoge?
17. Fault tolerance kya hoti hai?
18. Single point of failure (SPOF) kya hai, usse kaise avoid karein?
19. Disaster recovery strategy kya hoti hai?
20. Circuit breaker pattern kya hota hai?
21. Health checks kaise design karte ho system mein?
22. Failover mechanism kya hai?
23. Redundancy kaise achieve karte ho infrastructure mein?
24. Multi-region/Multi-AZ architecture kya hoti hai?
25. RTO aur RPO kya hote hain?

DevOps-Specific

26. CI/CD pipeline design karo ek microservices application ke liye.

27. Kubernetes architecture explain karo (Pods, Nodes, etcd, API server).
28. Kubernetes mein auto-scaling kaise kaam karta hai (HPA, VPA)?
29. Blue-Green deployment kya hoti hai?
30. Canary deployment kya hai, Blue-Green se kaise different hai?
31. Rolling update strategy kya hoti hai?
32. Zero-downtime deployment kaise design karoge?
33. Monitoring aur alerting system design karo (Prometheus + Grafana).
34. Log aggregation system kaise design karoge (ELK/EFK stack)?
35. Container orchestration ki zaroorat kyun padi?
36. Docker vs Kubernetes — role of each?
37. Infrastructure as Code (IaC) kya hai? (Terraform, Ansible)
38. Secrets management kaise karte ho production mein (Vault, KMS)?
39. Auto-scaling strategy design karo traffic spike ke liye.
40. Cost optimization strategies cloud infrastructure mein.

Practical / Scenario-Based

41. Design a URL shortener (jaise bit.ly).
42. Design a scalable logging system for a distributed application.
43. Design a system jo lakhs of concurrent users handle kare (e.g., ticket booking system).
44. Ek e-commerce website ke liye backend architecture design karo.
45. Design a notification system (email/SMS/push).
46. Design a rate limiter for an API.
47. Agar production mein sudden traffic spike aa jaaye, toh tumhara system kaise react karega? (Auto-scaling scenario)
48. Ek microservice down ho jaaye toh poora system kaise stable rahega? (Resilience scenario)
49. Design a CI/CD pipeline jisme deployment fail ho jaaye toh automatic rollback ho.
50. Design a monitoring system jo real-time alerts de agar server down ho ya CPU/memory usage high ho.

Tip: DevOps interview mein zyada focus 'DevOps-Specific' section ke questions (Kubernetes, CI/CD, monitoring, deployment strategies) pe hoga — inhe sabse zyada practice karo. Apne actual production project experience ko deeply explain karne ki practice bhi zaroor karo.